

Bits, Bytes & Units

Lesson 03 — Measuring Digital Information

JKSSB Computer Awareness

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Video Lesson 03

Watch: Bits, Bytes & KB-TB

Why Units Matter

- ▶ A file is reported as 4 MB; a network plan is reported as 20 Mbps.
- ▶ Both describe digital data, but one uses bytes and the other uses bits per second.
- ▶ Mixing the unit or the case of the letter can create an eightfold mistake.

The Small Building Blocks

- ▶ **Bit**: a binary digit, 0 or 1; the smallest unit of data.
- ▶ **Nibble**: 4 bits.
- ▶ **Byte**: 8 bits (two nibbles).
- ▶ **Word**: the natural data size a processor handles as a unit; its size depends on the architecture.
- ▶ **Character**: a text symbol; its encoded storage is not always one byte.

Case Is Part of the Unit

Do not confuse

b means bit; B means byte. One byte contains 8 bits.

- ▶ Mbps: megabits per second.
- ▶ MB: megabytes.
- ▶ At matching decimal prefixes, 8 Mbps corresponds to at most 1 MB/s before protocol overhead.

Capacity Ladder: Exam Convention

Conversion	JKSSB-style binary convention
1 KB	1024 bytes
1 MB	1024 KB
1 GB	1024 MB
1 TB	1024 GB

Use 1024 when an exam question uses the traditional KB/MB/GB/TB ladder. Modern standards also distinguish decimal SI units from binary IEC units such as KiB.

One-Step Conversion

Example

How many bytes are in 3 KB under the 1024 convention?

1. Write the given unit: 3 KB.
2. Use $1 \text{ KB} = 1024 \text{ bytes}$.
3. Calculate: $3 \times 1024 = 3072 \text{ bytes}$.

Check

State the convention and unit with the answer.

Capacity Is Not Speed

- ▶ **Capacity** tells how much data can be stored.
- ▶ **Transfer rate** tells how much data moves in a time interval.
- ▶ Storage is commonly labelled in bytes; data rates are commonly labelled in bits per second.
- ▶ A large-capacity drive is not automatically a fast drive.

Lesson 03: Recall

- ▶ 1 nibble = 4 bits; 1 byte = 8 bits.
- ▶ The bit/byte distinction depends on lowercase b vs uppercase B.
- ▶ A processor's word size is architecture-dependent.
- ▶ Under the exam's 1024 convention: $1 \text{ KB} = 1024 \text{ bytes}$.